

The Next Generation Sequencing Quality Initiative

The Next Generation Sequencing (NGS) Quality Initiative is a collaboration between the Centers for Disease Control and Prevention (CDC), the Association of Public Health Laboratories (APHL), and state and local public health laboratories (PHLs) to address the many challenges laboratories encounter when implementing NGS-based assays. The Initiative is developing an NGS-focused quality management system (QMS) to assure foundational quality during the development and implementation of sequencing-based tests by providing customizable, ready-to-implement tools and resources that laboratories can use to standardize and institute quality management practices and procedures. The NGS Quality Initiative has published additional tools and resources, including templates and procedures, that may be of assistance to laboratories throughout their NGS workflow. Please visit the following website to access these resources: <https://www.cdc.gov/labquality/qms-tools-and-resources.html>.

This document is intended to be used as a tool for implementing, improving, or maintaining an NGS QMS. Blue text provides examples for appropriate input and can be changed, deleted, or augmented as needed for the laboratory's specific requirements.

These documents and tools are not controlled files; format and content must be modified as needed to meet the document control, QMS, or regulatory requirements within your laboratory. It is the responsibility of your laboratory to take any necessary actions to ensure the information within these documents remains applicable.

Disclaimer:

Use of trade names and commercial sources is for identification only and does not imply endorsement by the U.S. Centers for Disease Control and Prevention or by the U.S. Department of Health and Human Services.

1. Scope

2. Method Validation Report for: (insert method name)
3. Branch/Laboratory: (branch and laboratory name)
4. Test System Name: (insert test system name and indicate whether new or existing test system)
5. Optional: Test Order Code (Client services manual): (Test code)

1. Test Procedure Document Number(s) and Revision Number:

2. Changes from the method validation plan, as applicable:

If new document versions were created, provide details describing the change. Examples include changes to the test method, changes to the number, or origin of samples. If new document versions were created, record them in the table.

1. Summary of the Test Procedure Purpose/Principle:

- 2.
- 3.
4. Assay result type (choose one):

Qualitative

Qualitative-titered

1. Specimen Condition Parameters:

- 2.
3. Information to support stability for submitted isolates: If applicable, the table below is used to reference related documentation for isolate stability.

4. Summary of Results:

- 5.
6. For minimum acceptable values, refer to acceptance criteria provided in section 6.0 in the Method Validation Plan; if values are not met, but the test process is revised, this should be detailed below in section 6.0.

TP- True Positive, TN- True Negative, FP- False Positive, FN- False Negative, CV- Coefficient of Variation

1. Interfering Substances:

2. If interference is observed during these studies, the interferent should be tested by serial dilutions to determine the lowest concentration that provides interference. Assay limitations may

be added to the validation summary.

1. Negative interferences:

1. Positive interferences:

2. Interpretation of Results:

3.

4. Evaluation of discrepant results: Provide details describing any discrepant results when compared to the gold standard or expected value. Provide evidence to support whether the new method is likely true or not true result, as needed. If a decision is made to recategorize/exclude the sample from validation, additional evidence needs to be provided to support this decision.

5. Limitations: Provide a description of method limitations. Significant cross-reactivities, interfering substances, and other conditions known to affect test results should be included. This may include patient preparation (timing of collection, antimicrobial therapy, acute/convalescent sample timing, etc.) and sample handling (e.g., time, temperature, etc.) Utilize the following tables for guidance as applicable. Statements may include: 1) The method is not appropriate to determine (describe the restriction). 2) The method is not appropriate for use under (describe the condition(s) under which it should not be used). 3) The method is not appropriate as a stand-alone test.

6. Mitigating Processes: Describe processes used to mitigate the assay limitations (e.g., reject certain specimen types, reflexive testing for certain results, assay always performed in conjunction with other assays).

7. Disclaimers (as applicable): Provide pertinent method limitations to be included on the final test report.

8. Fast Tracked test QA monitoring plan (as applicable): For tests that have been fast tracked due to urgent public health needs, describe the metrics to be monitored and the additional evidence of test performance characteristics to be gathered. Include the timeline for gathering this evidence.

9. Statement of Suitability:

10.

11. The method validation of (method name) has been completed according to the documented plan. The (method name) meets all the acceptance criteria and is approved for use in the (Insert name) Laboratory.

12. The method validation is applicable to the documents listed within the Method Validation Plan. Subsequent revisions involving technical changes to the procedure may require additional validation.

13. Responsibilities

14. Appendices

15.

16. Appendix A—Summary Data

17. Appendix B—Specimen Stability

1. Revision History

2. Approval

Approved By: Date:

Author

Print Name and Title

Approved By: Date:

Team Lead/Supervisor

Print Name and Title

Approved By: Date:

Quality Manager

Print Name

Appendix A—Summary Data

Please provide a line-by-line listing of all samples and results, and relevant summary data. Excel spreadsheets are acceptable.

1. Raw data: Please enter file path location of raw data here: _____.

NOTE: Please retain all validation data (worksheets, controls) and have available for review for the life of the procedure plus two years (CLIA).

1. Isolate stability Attestation form (as applicable)

Appendix B- Specimen Stability

1. Specimen Stability: The table below is used to document specimen stability study conditions and results for various conditions of specimen shipping, handling, and storage.